Frequency Response and Bode Plots

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Where this fits:

- **Builds on:** Second-order systems: natural frequency and damping ratio; Transfer functions.
- **Leads to:** Loop shaping: lead, lag and lead-lag; Cascade control; Filters in the loop; Aliasing and anti-aliasing filters.
- **Compare with:** Nyquist criterion.

1. Overview

Key point: A planner seldom asks for one turn; to weave between obstacles it swings the heading command back and forth. A linear loop responds to a sine with a sine at the same rate, only taller or shorter and later, so two numbers per weave rate, the gain and the phase, say how well the robot follows; the Bode plot draws them for every rate at once.

The robot's heading loop already turns the robot on command. Its PD controller (G-2962) pushes with a turning torque in proportion to the heading error and brakes in proportion to the turn rate. RO-060 tuned it to Kp = 0.2 N · m/rad and Kd = 0.283 N · m · s/rad, so that a turn just does not pass its target, and adds integral action (G-2963), Ki = 0.05, only against a dragging gearbox. The loop's transfer function (G-2999) holds its behaviour in one formula.

In this Note we study RO-061's lighter-damped first try instead: Kp = 0.4 N · m/rad and Kd = 0.08 N · m · s/rad, with no integral action. Its step response (G-2981) swings 53 percent past a 0.5 rad turn, to 0.763 rad. We take it because its light damping shows every effect of this Note, resonance included; Section 7 compares it with RO-060's tuned loop.

The planner does not, however, ask for one turn. To weave between obstacles, it swings the heading command left and right like a wave, a reference that keeps changing, which the loop must track (G-2954). Figure 1 sends the robot a weave of 0.1 rad (5.7°) to each side at three weave rates and shows its heading once the start-up swing has died away:

- a slow weave, 0.5 rad/s (one weave every 12.57 s): the heading **follows**, swinging 1.06 times as far as the command and 6° late;
- a weave at 1.92 rad/s (one every 3.27 s): the robot **over-turns**, swinging 2.55 times as far as the command and 78° late;
- a fast weave, 5 rad/s (one every 1.26 s): the robot **barely turns**, 0.19 times as far and 169° late, almost the opposite way to the command.

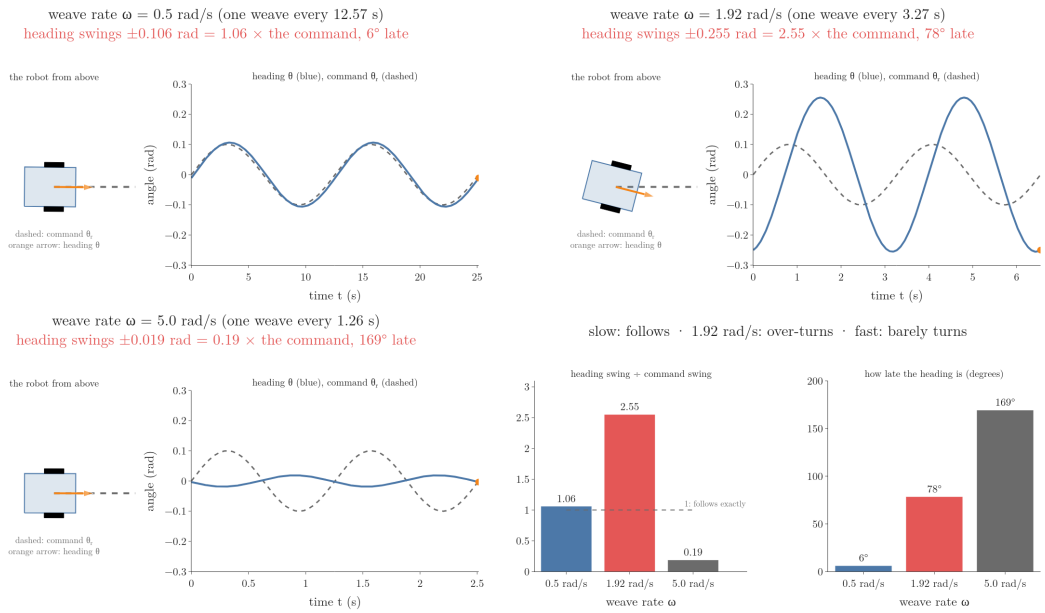


Figure 1: The robot from above (left) and its heading over two weave periods (right) while the heading command θ_r (dashed) swings 0.1 rad to each side. Each stage is one weave rate, shown after the start-up swing has died away: at 0.5 rad/s the heading swings ±0.106 rad, 1.06 times the command and 6° late; at 1.92 rad/s ±0.255 rad, 2.55 times and 78° late; at 5 rad/s ±0.019 rad, 0.19 times and 169° late. The last frame compares the three: heading swing divided by command swing, and how late the heading is in degrees

The same loop, the same 0.1 rad command, and three different responses. The opening question of this Note: **for a heading command that swings as a wave, how far does the robot turn and how late, at each weave rate, and which rates should the planner avoid?** We answer it in steps:

- why the heading is again a sine at the same rate, and the two numbers that describe it, the gain and the phase (Section 2);
- how the transfer function gives both numbers for any rate without a simulation (Section 3);
- the Bode plot, which draws them for every rate (Section 4);
- straight lines that sketch the plot by hand (Section 5);
- resonance, the reason for the over-turn at 1.92 rad/s (Section 6);
- bandwidth, the fastest weave the loop still follows (Section 7);
- where real autopilots measure and use these numbers (Section 8).

2. A sine in gives a sine out

> **Key point:** A loop built only from fixed multipliers, rates of change, running totals and sums turns a sine into a sine of the same rate. Only two things can differ: how far the output swings, the gain, and how late it is, the phase.

2.1 Why only the size and the timing change

In Figure 1 the heading never swings in some new shape: at each weave rate it is again a smooth wave at the command's own rate. Why should a weave command give a weave at all, and why at the same rate?

The loop is built from four simple operations, and each keeps a sine a sine of the same rate (FBS §6.3):

1. **Multiplying by a fixed number**, such as the gain Kp = 0.4: the wave gets taller or shorter.
2. **Taking the rate of change**, such as the turn rate the derivative action brakes with: the rate of change of a sine is the same wave a quarter period earlier.
3. **Taking the running total**, such as the heading, which is the running total of the turn rate: the running total of a sine swings as the same wave a quarter period later.
4. **Adding two signals**, such as the two torques of the PD controller: two waves of the same rate add up to one wave of that rate.

Figure 2 shows all four on the speed loop's PI controller: a drive force of 20 N per m/s times the speed error (its proportional gain), plus 10 N per m times the error's running total (its integral gain). We pick it because one controller then shows a multiplier, a running total and a sum at once. We feed it a speed error e, in m/s, that swings 0.1 m/s at the rate 0.5 rad/s, with t the time in s:

$$e = 0.1 \sin(0.5t)$$

Watch the last stage: the purple output is again a smooth wave of the same rate, taller and later.

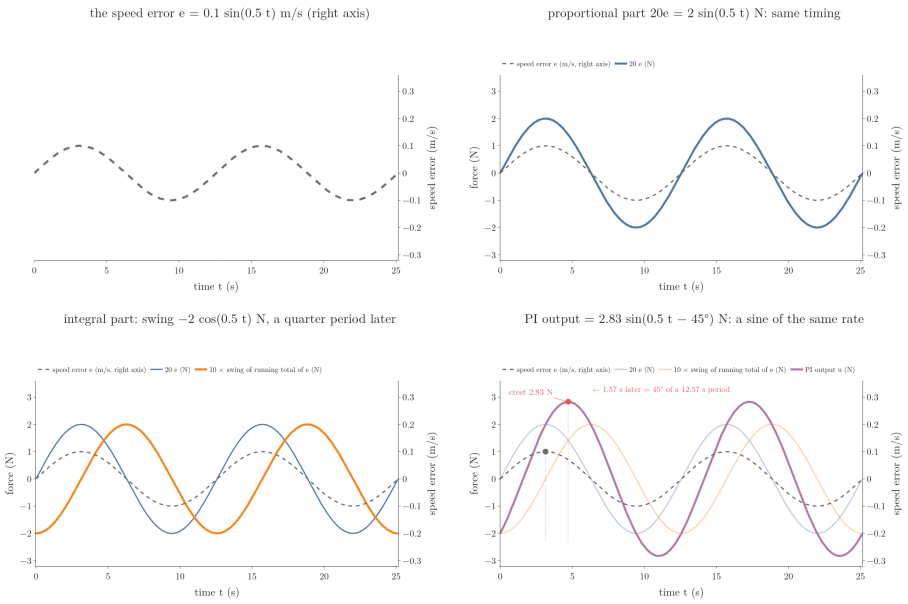


Figure 2: The speed loop's PI controller, force = 20 N per m/s × speed error + 10 N per m × running total of the error, fed the speed error e = 0.1 sin(0.5 t) m/s (grey, dashed, right axis). Stage 1: the error. Stage 2: the proportional part 20e = 2 sin(0.5 t) N (blue), crests at the same times as the error. Stage 3: the integral part, whose swing is −2 cos(0.5 t) N (orange), crests a quarter period (3.14 s) later. Stage 4: the sum (purple), 2.83 sin(0.5 t − 45°) N: a wave of the same rate, its crests 1.57 s after the error's

The sum, step by step. The proportional part is 20 times the error:

$$20 \times 0.1 \sin(0.5t) = 2 \sin(0.5t)$$

The running total of the error rises and falls about a fixed level; the fixed level is not a wave and we leave it out. Its swing, in m, is the wave below, whose rate of change is the error again:

$$-0.2 \cos(0.5t)$$

Times the integral gain, 10 N per m:

$$10 \times (-0.2 \cos(0.5t)) = -2 \cos(0.5t)$$

The two parts, both in N, add up to one wave:

$$2 \sin(0.5t) - 2 \cos(0.5t)$$

$$= 2.83 \sin(0.5t - 45°)$$

Why it holds. Any sum of a sine and a cosine of the same rate is one sine of that rate (Control System Lectures, Bode plots). With:

- a, the size of the sine part, here 2;
- b, the size of the cosine part, here -2;
- x, the angle inside both, here 0.5 t;
- φ, the shift, the angle of the point (a, b) from the positive x-axis, here $-45°$ for the point (2, -2).

$$a \sin x + b \cos x$$

$$= \sqrt{a^2 + b^2} \, \sin(x + \varphi)$$

With our numbers the size is:

$$\sqrt{2^2 + (-2)^2} = \sqrt{8} = 2.83$$

A check at an easy value. At the time 0, the left side:

$$2 \times 0 - 2 \times 1 = -2$$

The right side:

$$2.83 \times \sin(-45°)$$

$$= 2.83 \times (-0.707) = -2.0$$

The two sides agree. A shift of 45° is one eighth of a period, and one period of a 0.5 rad/s wave lasts 12.57 s, so the output's crests come later by:

$$12.57 \text{ s}/8 = 1.57 \text{ s}$$

The same four operations make up the heading loop, so a weave command must give a weave heading at the command's rate. One condition: the loop must stay linear. The torque must stay inside its limit of 0.2 N · m, because a clipped torque is no longer a fixed multiple of the error. In Figure 1 the largest torque is 0.094 N · m, at 1.92 rad/s (notebook). The sine in, sine out rule tells us the shape of the response, but not its size or its delay, which the next section measures.

2.2 Gain and phase: the frequency response

Figure 1 showed three weave rates; to know how far and how late the robot follows at any rate, we need two numbers per rate. Figure 3 measures them at one rate, 1 rad/s. The robot starts at rest and the command swings 0.1 rad to each side. Watch the shaded start-up part: the heading first swings at the loop's own rate, then settles onto a steady wave. The zoom of the last frame reads off the two numbers.

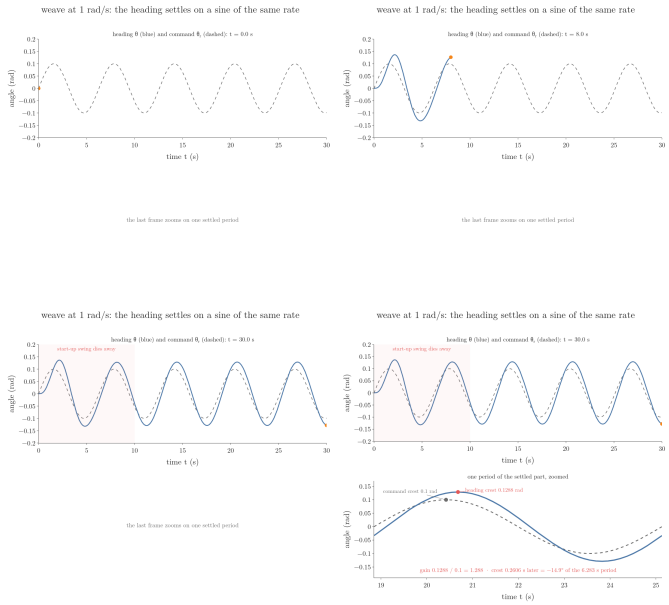


Figure 3: The heading command 0.1 sin(t), a weave at 1 rad/s (grey, dashed), and the heading of the PD loop (blue), started from rest, over 30 s. The start-up swing (shaded) dies away by about 10 s; after that the heading is a sine of the same rate. The last frame zooms on one period from 18.85 s: the heading's crest is 0.1288 rad against the command's 0.1 rad, and it comes 0.2606 s later, which is −14.9° of the 6.283 s period

Why the start-up part dies away. At the start the robot points straight ahead while the command points elsewhere; the gap sets off the loop's own swing. That swing shrinks because the loop is stable: its poles (G-3000) sit at -0.4 ± 1.96j, and the real part -0.4 sets how fast the swing's size shrinks. After 10 s it is this fraction of its start (notebook):

$$e^{-0.4\times 10} = 0.018$$

What is left, the settled part or steady state, depends only on the command. The two numbers below therefore describe the loop, not the way it started (FBS §6.3).

The two numbers. With:

- ω, the weave rate, 1 rad/s here;
- T_w, one period of the weave, 6.283 s (2π divided by the rate);
- A_r, how far the command swings, 0.1 rad;

- A, how far the heading swings once settled, 0.1288 rad;
- Δt, how much later the heading's crest comes than the command's, 0.2606 s.

The ratio of the two swings is the **gain** (G-3021) at that rate (FBS §6.3), written M:

$$M = A/A_r$$

$$= 0.1288/0.1 = 1.288$$

The delay, as a fraction of a period and turned into degrees, is the **phase** (G-3022) at that rate, written φ; it is negative when the output is late (FBS §6.3):

$$\varphi = -360^\circ \times \Delta t/T_w$$

$$= -360^\circ \times 0.2606/6.283$$

$$= -14.9^\circ$$

The gain and the phase, given for every rate ω, are the loop's **frequency response** (G-3020) (FBS §6.3; FBS §9.6). In Figure 1 the frequency response at three rates gave three pairs: (1.06, −6°), (2.55, −78°) and (0.19, −169°). Measuring each rate with a simulation, as here, takes 30 s of simulated driving per rate. The transfer function gives the same two numbers with one line of algebra.

3. Gain and phase from the transfer function

> **Key point:** Replacing the transfer function's variable s by jω, where ω is the weave rate and j squares to −1, gives one complex number for each weave rate: its length is the gain and its angle is the phase.

3.1 Putting the weave rate into the transfer function

To answer the opening question at every weave rate, we cannot simulate every rate. The loop's equation gives the gain and phase directly.

Two recaps.

- **Complex numbers.** The number j squares to −1. A complex number such as 3 + 4j is an arrow on a plane, 3 across and 4 up; its length is 5, by Pythagoras, and its angle from the positive real axis is 53.1°:

$$\sqrt{3^2 + 4^2} = 5$$

- **A spinning arrow.** By Euler's formula, the arrow below has length 1 and turns at ω radians per second, so its upward part is a sine of rate ω:

$$e^{j\omega t} = \cos\omega t + j\sin\omega t$$

Its rate of change is $j\omega$ times itself. For this arrow, taking a rate of change therefore becomes multiplying by $j\omega$. Multiplying by $j\omega$ is the transfer function's own idea, an exponential e^{st} in and the same exponential out, with $s = j\omega$.

The loop's equation. The yaw inertia I = 0.1 kg · m^2 turns under the PD torque, Kp times the heading error minus Kd times the turn rate (the derivative acts on the measured turn rate, as in RO-061). With:

- θ, the heading, in rad, and θ', θ'', its rate of change and that rate's rate of change;
- θ_r, the heading command, in rad;
- I, the yaw inertia, 0.1 kg · m^2;
- K_p, the proportional gain, 0.4 N · m/rad;
- K_d, the derivative gain, 0.08 N · m · s/rad.

$$I\theta'' = K_p(\theta_r - \theta) - K_d\theta'$$

$$0.1\theta'' + 0.08\theta' + 0.4\theta = 0.4\theta_r$$

The derivation. We feed the loop the spinning arrow e^{jt}, a weave at 1 rad/s, and guess that the heading is the same arrow times some complex number Y. Our numbers first:

$$\theta_r = e^{jt}$$

$$\theta = Ye^{jt}$$

$$\theta' = jYe^{jt}$$

$$\theta'' = -Ye^{jt}$$

$$(-0.1 + 0.08j + 0.4)\,Ye^{jt}$$

$$= 0.4\,e^{jt}$$

$$(0.3 + 0.08j)\, Y = 0.4$$

$$Y = 0.4/(0.3 + 0.08j)$$

The same steps for any weave rate ω:

$$\theta_r = e^{j\omega t}$$

$$\theta = Y e^{j\omega t}$$

$$\theta' = j\omega Y e^{j\omega t}$$

$$\theta'' = -\omega^2 Y e^{j\omega t}$$

$$(-I\omega^2 + j\omega K_d + K_p)\, Y e^{j\omega t}$$

$$= K_p\, e^{j\omega t}$$

$$(-I\omega^2 + j\omega K_d + K_p)\, Y = K_p$$

$$Y = \frac{K_p}{K_p - I\omega^2 + j\omega K_d}$$

$$= G(j\omega)$$

The last line is the loop's transfer function with s replaced by $j\omega$:

$$G(s) = \frac{K_p}{Is^2 + K_d s + K_p}$$

The real command, a sine of 0.1 rad, is the upward part of the spinning arrow times 0.1. The heading is therefore the upward part of that arrow times 0.1 and times Y: a sine whose size is 0.1 times the length of Y and whose shift is the angle of Y. For $\omega = 1$ rad/s:

$$|Y| = 0.4/\sqrt{0.3^2 + 0.08^2}$$

$$= 0.4/0.3105 = 1.288$$

$$\angle Y = -\arctan(0.08/0.3) = -14.9^\circ$$

These are the gain and the phase that Figure 3 measured. The frequency response, the gain and phase at each weave rate, is therefore the transfer function at s = jω: the gain is the length of G(jω) and the phase is its angle (FBS §9.6). One complex number per rate replaces 30 s of simulation, but one number at a time does not yet show how the response changes with the rate.

3.2 The frequency response as an arrow

Section 3.1 gave one arrow for the weave at 1 rad/s: length 1.288 at an angle of -14.9°. As the weave rate rises, the arrow changes, and its changes are the three responses of Figure 1. Figure 4 sweeps the rate from 0.1 to 10 rad/s. Watch the orange arrow's length (the gain) and its angle from the positive real axis (the phase, red arc).

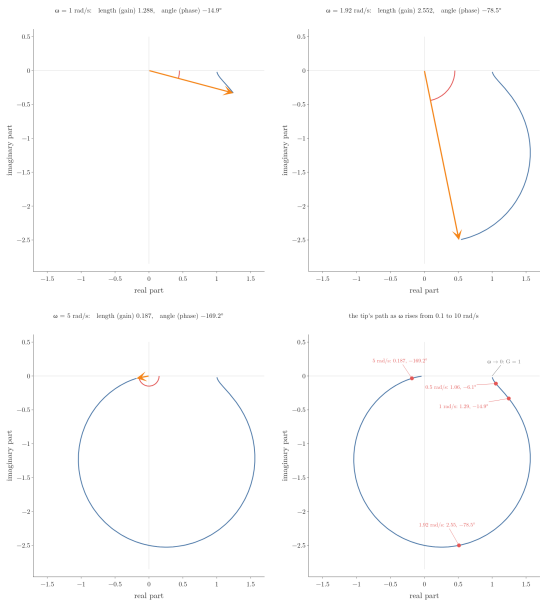


Figure 4: The complex number G(jω) of the heading loop drawn as an orange arrow from 0, its angle marked by a red arc, while the weave rate ω rises from 0.1 to 10 rad/s; the tip leaves a blue trail. At 1 rad/s the arrow has length 1.288 and angle -14.9°; at 1.92 rad/s it is long, 2.55, and points almost straight down, -78.5°; at 5 rad/s it is short, 0.187, and points almost left, -169.2°. The last frame shows the whole trail with these rates marked; as ω goes to 0 the tip sits at 1

The arrow summarises Figure 1:

- at slow rates it has length about 1 and points right: the heading **follows** the command;

- near 2 rad/s it grows long and turns to point down, $-90°$: the robot **over-turns** and is a quarter period late;
- at fast rates it shrinks and turns to point left, $-180°$: the robot **barely turns**, and the little it turns is opposite to the command.

One picture holds the whole frequency response, but the rates are hidden along the trail: to read the gain at 3 rad/s we need a label at every point. The next section draws the gain and the phase each against the rate.

4. The Bode plot

> **Key point:** The Bode plot draws the gain in decibels and the phase in degrees against the weave rate on a log scale; decibels turn the products of a loop's pieces into sums, and the log scale gives slow and fast rates the same room.

4.1 Decibels and a log rate axis

To read the robot's response at any weave rate, we want the gain and the phase as two curves against the rate. Two problems appear on plain axes. The heading loop's gain runs from 2.55 at 1.92 rad/s down to 0.01 at 20 rad/s, so on a plain scale every gain below 0.1 is squeezed onto the bottom line. And the rates of interest run from 0.1 to 20 rad/s, so on a plain scale everything below 1 rad/s is squeezed into the left edge.

Decibels. We write a gain in decibels (dB), 20 times its base-10 logarithm, the unit used for sensor signals. With M, the gain (the output's swing divided by the input's), a number without a unit:

$$\text{gain in dB} = 20 \log_{10} M$$

Some values:

- a gain of 10 is 20 dB, and a gain of 0.1 is -20 dB;
- a gain of 1, output as large as input, is 0 dB;
- a gain of 0.707 is -3.01 dB;
- the resonant gain 2.552 is 8.14 dB.

The factor is 20, not 10, because a decibel compares powers, and a wave's power grows with the square of its size (Control System Lectures, Bode plots). Decibels also turn products into sums: the logarithm of a product is the sum of the logarithms. A loop made of pieces multiplied together, such as a controller times a plant, has a gain in dB that is the sum of the pieces' gains in dB, and a phase that is the sum of their phases (FBS §9.6). Section 5 builds on that.

A log rate axis. On the rate axis we give each factor of 10 the same width: 0.1 to 1 rad/s takes as much room as 1 to 10 rad/s. A factor of 10 in rate is a **decade** (G-3024). With gains in decibels and rates on a log axis, the robot's responses at all rates fit on one page.

4.2 Building the plot one weave rate at a time

With these two axes we can now plot the robot's response at many weave rates. Figure 5 repeats the measurement of Figure 3 at twelve rates from 0.2 to 10 rad/s. Each frame shows the settled command and heading for one rate (top), then adds one dot to the gain plot (middle) and one to the phase plot (bottom). Watch the dots climb to a peak near 2 rad/s and then fall away while the phase slides from 0° to −180°.

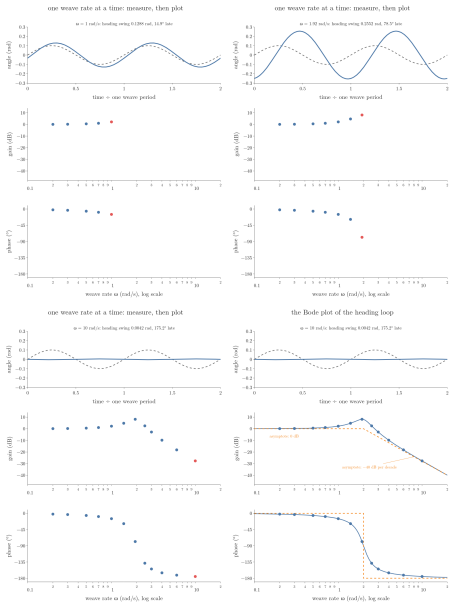


Figure 5: One weave rate per frame: two periods of the settled command (grey, dashed) and heading (blue), then the gain in decibels (middle) and the phase in degrees (bottom), one red dot per rate, from 0.2 to 10 rad/s on a log scale. The dots come from simulating the loop from rest at each rate and fitting a sine to the settled heading. The last frame draws the exact curves of G(jω) through the dots and the straight-line asymptotes (orange, dashed): 0 dB up to 2 rad/s, then −40 dB per decade; phase 0°, then −180°

The pair of plots, gain in dB and phase in degrees against the rate on a log scale, is the **Bode plot** (G-3023) (FBS §9.6). The heading loop's Bode plot has three regions, the three responses of Figure 1:

- **below about 1 rad/s** the gain is near 0 dB and the phase near 0°: the robot follows;
- **near 2 rad/s** the gain peaks at 8.14 dB, a gain of 2.55, and the phase passes −90°: the robot over-turns;
- **above about 3 rad/s** the gain falls 40 dB for every decade and the phase approaches −180°: the robot barely turns, and the wrong way.

Each dot took a simulation; the exact curve took the formula of Section 3.1 at 600 rates. A designer without a computer, or one who wants to see why the curve bends where it does, needs a third way: the straight lines drawn in orange.

5. Sketching a Bode plot from straight lines

> **Key point:** Far below and far above each corner, a loop's gain in dB lies on straight lines, flat or falling 20 dB per decade for each pole; the lines meet at the corner frequency, where the real curve departs from them most.

5.1 One pole: the speed plant

To see why the heading loop's gain is flat and then falls, we start with the simplest piece on the robot: the speed plant, which has a single pole. The drive force pushes the 8 kg robot against a rolling drag of 2 N · s/m. Its transfer function splits into two pieces, a fixed number and a lag. With:

- $P(s)$, the speed plant's transfer function, from drive force (N) to speed (m/s);
- s, the transfer function's variable: for an exponential input e^{st}, taking a rate of change multiplies by s.

$$P(s) = \frac{1}{8s + 2}$$

$$= 0.5 \times \frac{1}{4s + 1}$$

Here 0.5 m/s per newton is the speed one newton holds against the drag (the fixed piece, which we call K), and 4 s is the plant's time constant (G-2988), the time it takes to close 63 percent of a gap: here the mass divided by the drag. Figure 6 builds its Bode plot (gain in decibels and phase against the rate) piece by piece. Watch stage 3: in decibels the two pieces add.

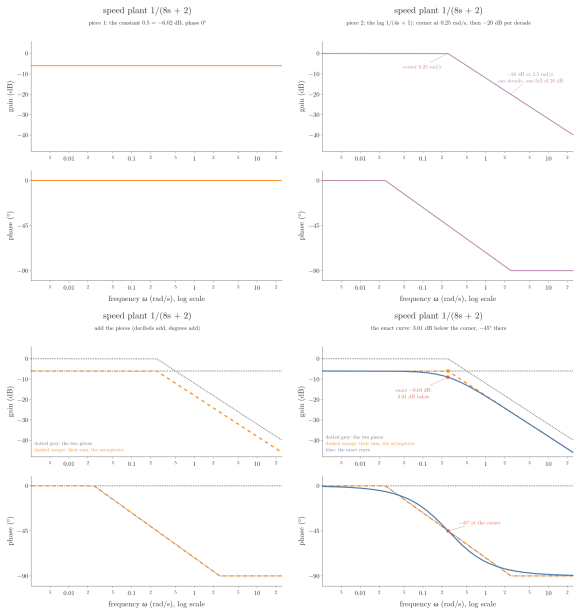


Figure 6: The Bode plot of the speed plant $1/(8s + 2)$ built from pieces, rate on a log scale from 0.0025 to 25 rad/s. Stage 1: the constant 0.5 is −6.02 dB at every rate, phase 0°. Stage 2: the lag $1/(4s + 1)$ is flat at 0 dB up to its corner 0.25 rad/s, then falls 20 dB per decade; its phase falls in a straight line from 0° at 0.025 rad/s to −90° at 2.5 rad/s. Stage 3: the pieces (grey, dotted) add to the asymptotes (orange, dashed): −6.02 dB up to 0.25 rad/s, then −20 dB per decade. Stage 4: the exact curve (blue) is −9.03 dB at the corner, 3.01 dB below the asymptotes, with phase −45° there

Why the gain is flat, then falls. With:

- ω, the rate of the force wave, in rad/s;
- T, the time constant, 4 s (the symbol RO-061 uses).

The lag piece at $s = j\omega$ is:

$$\frac{1}{1 + j\omega T} = \frac{1}{1 + 4j\omega}$$

Two easy cases:

- **Slow waves**, ω much smaller than 0.25 rad/s: the term $4j\omega$ is small compared with 1, so the piece is close to 1, which is 0 dB with phase 0°. The robot's speed keeps up with a slow push.
- **Fast waves**, ω much larger than 0.25 rad/s: the 1 is small compared with $4j\omega$, so the piece is close to the line below. Ten times the rate gives a tenth of the gain, which is −20 dB per decade, and dividing by j is a phase of −90°. The 8 kg mass cannot change speed fast enough to follow a fast push.

$$\frac{1}{1 + 4j\omega} \approx \frac{1}{4j\omega}$$

The two straight lines are the plot's **asymptotes** (G-3025), the lines the gain curve approaches far from the bend (FBS §9.6). They meet at the rate where the two terms are equal in size, one over the time constant:

$$\omega = 1/T = 1/4 = 0.25 \text{ rad/s}$$

That rate is the **corner frequency** (G-3026).

At the corner itself, the real curve is furthest from the lines (MATLAB, Understanding Bode plots 3). Our numbers first:

$$P(0.25j) = \frac{0.5}{1+j}$$

$$|1+j| = \sqrt{2} = 1.414$$

$$|P| = 0.5/1.414 = 0.3536$$

$$\angle P = -\arctan(1) = -45^\circ$$

The same in symbols, with K the fixed piece, 0.5 m/s per N, and T the time constant, 4 s:

$$P(j/T) = \frac{K}{1+j}$$

$$|P| = K/\sqrt{2}$$

$$\angle P = -45^\circ$$

In decibels the two pieces add, the fixed 0.5 and the lag's 0.707 at its corner:

$$-6.02 - 3.01 = -9.03 \text{ dB}$$

$$20 \log_{10} 0.3536 = -9.03 \text{ dB}$$

So a force wave of 1 N at 0.25 rad/s makes the speed swing 0.3536 m/s, 45° late. One period lasts 25.13 s, and 45° is an eighth of it:

$$25.13 \text{ s}/8 = 3.14 \text{ s late}$$

The phase line. The phase is close to 0° up to a tenth of the corner, 0.025 rad/s (exact −5.7°), and close to −90° from ten times the corner, 2.5 rad/s (exact −84.3°). In between, a straight line falling 45° per decade passes −45° at the corner (FBS §9.6). One pole gives one bend; the heading loop has two.

5.2 Two poles: the heading loop

The heading loop's straight lines in Figure 5 follow from the same two easy cases. We write its transfer function in the standard second-order form, with:

- ω_n, the natural frequency (G-2991), the rate of the loop's own swing, 2 rad/s for Kp = 0.4 and I = 0.1;
- ζ, the damping ratio (G-2992), how strongly the derivative action brakes that swing, 0.2 for Kd = 0.08.

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{4}{s^2 + 0.8s + 4}$$

At s = jω the term s^2 becomes:

$$s^2 = (j\omega)^2 = -\omega^2$$

Two easy cases:

- **Slow weaves**, ω much smaller than 2 rad/s: the 4 in the denominator dominates, so G is close to 4 divided by 4, which is 1: 0 dB with phase 0°.
- **Fast weaves**, ω much larger than 2 rad/s: the $-\omega^2$ term dominates, so G is close to the line below. Ten times the rate gives a hundredth of the gain, which is −40 dB per decade, and the minus sign is a phase of −180°: the heading swings opposite to the command.

$$G(j\omega) \approx -4/\omega^2$$

The check at 20 rad/s: the asymptote gives −40 dB and the exact gain is −39.92 dB (notebook). Each pole adds −20 dB per decade above its corner and up to −90° of phase, so two poles give twice both (FBS §9.6).

Near the corner ω_n = 2 rad/s the straight lines say 0 dB, but the exact curve in Figure 5 stands 8 dB above them. The lines miss a peak, and how far they miss depends on the damping ratio ζ (FBS §9.6). That peak is the over-turn of Figure 1.

6. Resonance: the weave that over-turns the robot

> **Key point:** A lightly damped loop has a weave rate at which each push lands in step with the loop's own swing; there the output swings far more than the input. For our loop the peak is 2.55 times the command at 1.918 rad/s, close to the 2.5 that one over twice the damping ratio $\zeta = 0.2$ gives at the natural frequency; with ζ of 0.707 or more there is no peak.

At 1.92 rad/s the robot over-turns: it swings 2.55 times as far as the command asks. Why there, and why so much?

The idea in plain words. A child on a swing goes highest when each push comes in step with the swing. The heading loop is such a swing: the proportional torque pulls the heading back towards the command like a spring, the yaw inertia keeps it moving past, and the loop's own swing has the rate $\omega_n = 2$ rad/s. A weave command near that rate pushes each swing a little further. Only the derivative action, which brakes with the turn rate, stops the swing from growing without end. A large output at the rate of a system's own swing is **resonance** (G-3027) (FBS §6.3).

A check at the natural frequency. At $\omega = \omega_n = 2$ rad/s the two real parts of the denominator cancel, our numbers first:

$$G(2j) = \frac{4}{4 - 4 + 0.8 \times 2j}$$

$$= \frac{4}{1.6j} = 2.5 \angle -90^\circ$$

The same in symbols, using the form of Section 5.2:

$$G(j\omega_n) = \frac{\omega_n^2}{2\zeta\omega_n^2 j}$$

$$= \frac{1}{2\zeta} \angle -90^\circ$$

With $\zeta = 0.2$:

$$1/(2 \times 0.2) = 2.5$$

At its natural frequency the loop multiplies the command by 2.5 and is a quarter period late (FBS §9.6). Only the damping term is left in the denominator, so the smaller the damping ratio, the taller the swing.

Where the peak is exactly. The largest gain is the **resonant peak** (G-3028) and the rate where it occurs is the **resonant frequency** (G-3028) (FBS §6.3). The gain is largest where

the size of the denominator is smallest, and the lowest point of a smooth curve is where its slope is zero. With:

- x, the square of the weave rate ω, in rad^2/s^2;
- $D(x)$, the squared size of the denominator of $\text{G}(\text{j}\omega)$;
- ω_r, the resonant frequency, in rad/s;
- M_r, the resonant peak, the largest gain.

Our numbers first:

$$D(x) = (4 - x)^2 + 0.64x$$

$$\frac{dD}{dx} = -2(4 - x) + 0.64$$

$$0 = -8 + 2x + 0.64$$

$$x = 3.68$$

$$\omega_r = \sqrt{3.68} = 1.918 \text{ rad/s}$$

$$D = 0.32^2 + 0.64 \times 3.68 = 2.4576$$

$$M_r = 4/\sqrt{2.4576} = 2.552$$

The same in symbols:

$$D(x) = (\omega_n^2 - x)^2 + 4\zeta^2\omega_n^2 x$$

$$\frac{dD}{dx} = -2(\omega_n^2 - x) + 4\zeta^2\omega_n^2$$

$$x = \omega_n^2(1 - 2\zeta^2)$$

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2}$$

$$D = 4\zeta^4\omega_n^4 + 4\zeta^2\omega_n^4(1 - 2\zeta^2)$$

$$= 4\zeta^2\omega_n^4(1-\zeta^2)$$

$$M_r = \frac{\omega_n^2}{\sqrt{D}} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

The true peak, 2.552 (8.14 dB) at 1.918 rad/s, is therefore slightly taller than 2.5 and slightly below ω_n. The formula for x also says when there is a peak: only while x is positive, which needs:

$$1 - 2\zeta^2 > 0$$

$$\zeta < 0.707$$

From $\zeta = 0.707$ up, the gain falls from 0 dB at every rate.

Figure 7 redraws the gain curve as ζ falls from 1 to 0.1, by lowering Kd from 0.4 to 0.04 while Kp stays 0.4, so the natural frequency stays 2 rad/s. Watch the orange dot, $1/(2\zeta)$ at ω_n, and the red peak just left of it.

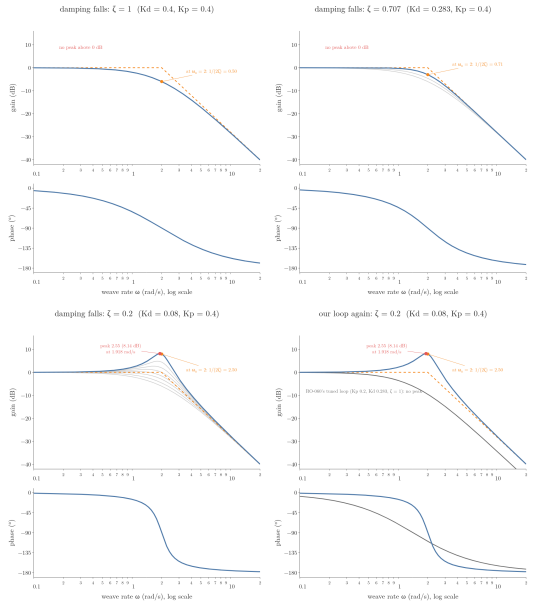


Figure 7: The heading loop's gain (top) and phase (bottom) with Kp = 0.4 while the damping ratio ζ falls from 1 (Kd = 0.4) to 0.1 (Kd = 0.04); earlier curves stay in light grey, the asymptotes are orange and dashed. For $\zeta = 1$, 0.85 and 0.707 the gain never rises above 0 dB. Below 0.707 a red peak grows just left of the natural frequency ωn = 2 rad/s, close to the orange dot $1/(2\zeta)$ at ωn; the phase drops from 0° to −180° more and more steeply. The last frame returns to our loop, $\zeta = 0.2$: peak 2.55 (8.14 dB) at 1.918 rad/s, against 2.5 at 2 rad/s, with RO-060's tuned loop (Kp 0.2, Kd 0.283, $\zeta = 1$) in grey

Figure 7 shows three things:

- **The peak and the overshoot have one cause.** The small damping ratio that lets the 0.5 rad turn overshoot by 53 percent in RO-061 is the one that lets a weave at 1.92 rad/s over-turn by a factor of 2.55.
- **RO-060's tuned loop has no peak.** Its gains, Kp = 0.2 and Kd = 0.283, give a damping ratio of 1 (at a natural frequency of 1.41 rad/s), and its gain never rises above 0 dB: the derivative chosen so a turn just does not pass its target leaves no weave rate amplified either.
- **$\zeta = 0.707$ is not critical damping.** It is the smallest damping ratio with no resonant peak, the one whose gain at ω_n, $1/(2\zeta)$, is 0.707, that is -3 dB. Critical damping, the smallest damping with no overshoot in a step response, is $\zeta = 1$, the critically damped (G-2995) case. Between the two, a step still overshoots slightly but no weave rate is amplified.

The planner should therefore keep the heading command's weave rates away from 1.9 rad/s, a weave period near 3.3 s, while the loop runs with $\zeta = 0.2$. Resonance explains the over-turn; the third answer of Figure 1, barely turning at 5 rad/s, needs a limit on the other side.

7. Bandwidth: how fast a weave the loop follows

> **Key point:** The bandwidth is the rate at which the gain has fallen to 0.707 of its zero-frequency value, -3 dB. Our heading loop's bandwidth is 3.02 rad/s; weaves faster than that get less than 71 percent of their size and come more than 150° late.

At 5 rad/s the robot barely turns. How fast a weave can the planner ask for before the robot stops following, that is before the gain, the heading's swing divided by the command's, drops away?

The idea. Above some rate, the robot's inertia dominates and its swing shrinks fast (Section 5.2: 40 dB less for every decade). A single number marks where following gives way: the rate at which the gain has fallen to 0.707 (one over the square root of 2) of its value for very slow commands, the zero-frequency gain. That rate is the **bandwidth** (G-3029) of the loop (FBS §6.3). The value 0.707 is a convention with a reason: a wave's power grows with the square of its size (Section 4.1), and 0.707 squared is 0.5, so at the bandwidth the heading's swing carries half the power it carries for a slow weave. In decibels, 0.707 is 3.01 dB below the zero-frequency gain, so the bandwidth is often called the -3 dB point.

Figure 8 marks the bandwidth on two gain curves. Look where each curve crosses the red dashed line.

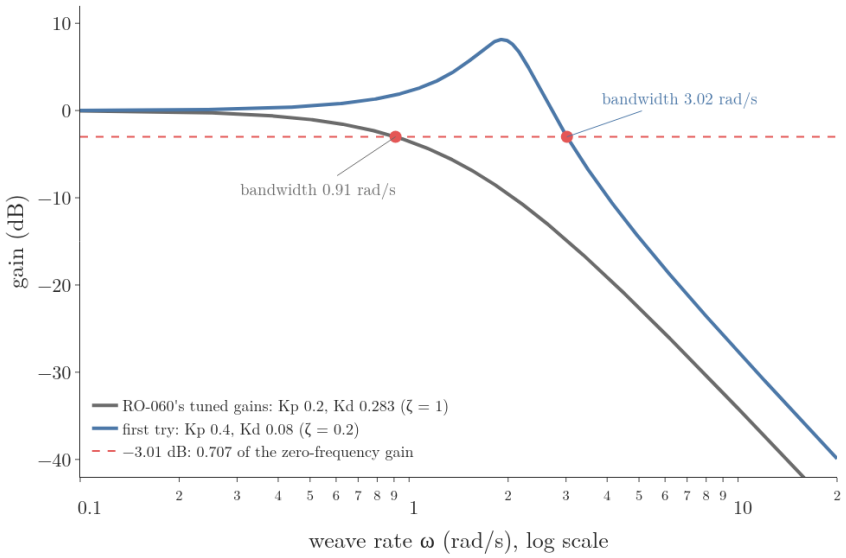


Figure 8: The gain in decibels of the heading loop with the first-try gains (Kp 0.4, Kd 0.08, $\zeta = 0.2$, blue) and with RO-060's tuned gains (Kp 0.2, Kd 0.283, $\zeta = 1$, grey), rate on a log scale. Both start at 0 dB for slow weaves. The red dashed line is −3.01 dB, 0.707 of the zero-frequency gain; the blue curve crosses it at its bandwidth 3.02 rad/s, after its resonant peak, and the grey curve at 0.91 rad/s

What the numbers mean for the robot:

- **Our loop: 3.02 rad/s**, a weave period of 2.08 s. Faster weaves, such as the 5 rad/s of Figure 1, get less than 71 percent of their size; at the bandwidth itself the heading is already 154.7° late (notebook).
- **Below the bandwidth is not the same as faithful.** Between 1 and 3 rad/s our loop's gain lies above 1: it over-turns there, as Section 6 showed. The resonant peak is a separate specification next to the bandwidth (FBS §12.2).
- **RO-060's tuned loop has a bandwidth of 0.91 rad/s.** Removing the overshoot, and halving Kp, costs speed: the tuned loop follows only slower weaves, but faithfully, with no rate over-turned.
- **The speed plant's bandwidth is its corner frequency, 0.25 rad/s,** because Section 5.1 showed the gain there is 0.707 of its zero-frequency value: force waves much faster than one every 25 s barely change the robot's speed.

A rule of thumb for design is a bandwidth 5 to 10 times the fastest rate the loop must follow (MATLAB, Using Bode plots 1). With 3.02 rad/s, our heading loop follows weaves up to about 0.3 to 0.6 rad/s well, which is what Figure 1 showed at 0.5 rad/s: 1.06 times the command and 6° late. Bandwidth completes the answer to the opening question; the next section shows where real autopilots measure frequency responses and guard against resonance.

8. Where frequency responses are measured and used

> **Key point:** Autopilots measure frequency responses with sine sweeps, filter sensor signals above the rates their loops need, and cut out structural resonances with notch filters.

The weave of Figure 5, a sine at rising rates, is how real vehicles are measured. Three examples from the PX4 autopilot, an open-source flight stack for drones; we read its parameter files on GitHub:

- **Sine sweeps.** PX4's fixed-wing auto-tuner shakes each axis of the aircraft with a sine whose rate sweeps from 1 Hz to 10 Hz over 10 s, a linear sweep by default (parameters `FW_AT_SYSID_F0`, `FW_AT_SYSID_F1`, `FW_AT_SYSID_TIME` and `FW_AT_SYSID_TYPE` in `src/modules/fw_autotune_attitude_control/fw_autotune_attitude_control_params.yaml`). One hertz is one cycle per second, 6.28 rad/s, so the sweep covers 6.28 to 62.8 rad/s. From the measured responses the tuner fits a model of the axis and computes new gains (`fw_autotune_attitude_control.cpp`).
- **Bandwidth of a filter.** The gyroscope signal that feeds PX4's turn-rate controller first passes a second-order low-pass filter whose cutoff, the −3 dB rate, is 40 Hz by default (`IMU_GYRO_CUTOFF` in `src/modules/sensors/vehicle_angular_velocity/imu_gyro_parameters.`. Slow turns pass almost unchanged; vibration from the motors, faster than 40 Hz, is cut.
- **Resonance.** The same file offers notch filters (`IMU_GYRO_NF0_FRQ`, `IMU_GYRO_NF0_BW`), which cut a narrow band of rates "to avoid feedback amplification of structural resonances": a frame that bends at its own rate has a peak like the one in Section 6, and the loop must not push it.

The frequency response has told us how a stable loop responds to each weave. The closed loop's frequency response $G(j\omega)$ cannot tell whether a loop is stable at all, for instance when the heading reaches the controller late; the Nyquist criterion answers that from the frequency response of the loop cut open, the controller times the robot.

9. Summary

The symbols used below:

- ω is the weave rate, in rad/s;
- $G(j\omega)$ is the heading loop's transfer function at s = jω, for our gains the complex number below;
- M is the gain, the heading's swing divided by the command's;
- φ is the phase, the heading's delay as a fraction of a period, in degrees;
- $\omega_n = 2$ rad/s and $\zeta = 0.2$ are the natural frequency and damping ratio of the loop.

$$G(j\omega) = \frac{4}{4 - \omega^2 + 0.8j\omega}$$

Idea	What it says	Why it matters
Sine in, sine out	a linear loop responds to a sine with a sine of the same rate	two numbers per rate describe the whole response
Frequency response	gain M and phase φ at every rate, equal to the length and angle of $G(j\omega)$	computed from the transfer function, no simulation needed
Bode plot	gain in dB and phase in degrees against the rate on a log scale	shows slow and fast rates together; pieces add
Asymptotes	flat, then -20 dB per decade per pole, meeting at the corner	sketches the plot by hand and shows where it bends
Resonance	a peak near ω_n of about $1/(2\zeta)$, none for $\zeta \geq 0.707$	the rates where the loop over-turns
Bandwidth	the rate where the gain falls to 0.707 (-3 dB)	the fastest command the loop still follows

- With Kp = 0.4 and Kd = 0.08 the robot follows slow weaves (0.5 rad/s: 1.06 times the command, 6° late), over-turns near 1.92 rad/s (2.55 times, 78° late) and barely turns above its bandwidth of 3.02 rad/s (5 rad/s: 0.19 times, 169° late), so the planner should keep weave rates below about 0.6 rad/s, a fifth of the bandwidth, or the loop needs more damping.
- A sine stays a sine of the same rate, because the loop only multiplies, takes rates of change and running totals, and adds, as long as the torque stays inside its 0.2 N · m limit.
- The gain and the phase are the length and the angle of G(jω), because a spinning arrow passes through the loop's equation multiplied by that complex number.
- The Bode plot uses decibels and a log rate axis, so a loop's pieces add and every rate from 0.1 to 20 rad/s gets room.

- A pole's asymptotes are flat and then fall 20 dB per decade from its corner, where the exact curve is 3.01 dB lower and the phase is $-45°$, so a sketch takes a few straight lines.
- The heading loop resonates because its proportional torque acts like a spring against the yaw inertia; the peak, 2.552 at 1.918 rad/s, grows as the damping ratio falls, so the same small ζ that made the 0.5 rad turn overshoot by 53 percent makes weaves near 1.92 rad/s over-turn.
- RO-060's tuned gains (Kp 0.2, Kd 0.283, $\zeta = 1$) have no peak but a bandwidth of 0.91 rad/s against 3.02, so damping is traded for speed.
- PX4 measures frequency responses with a 1 to 10 Hz sine sweep and guards its loops with a 40 Hz low-pass filter and notch filters, so the ideas here run on real autopilots.

We can now predict how the heading loop responds to any weave from $G(j\omega)$ alone. One question remains: whether a loop stays stable when its readings arrive late or its gains change, and by how much; the Nyquist criterion and the stability margins answer that from the frequency response of the loop cut open.

10. Sources

Built from

- Control System Lectures, "Bode Plots, Introduction", YouTube, https://www.youtube.com/watch?v=_ (Control System Lectures, Bode plots)
- MATLAB, "Understanding Bode Plots, Part 3: Exploring Bode Plots for Simple Systems", https://www.mathworks.com/videos/understanding-bode-plots-simple-systems-3-of-4-76213.html (MATLAB, Understanding Bode plots 3)
- MATLAB, "Understanding Bode Plots, Part 4: Exploring Bode Plots for Higher-Order Systems", https://www.mathworks.com/videos/understanding-bode-plots-complex-systems-4-of-4-76214.html (MATLAB, Understanding Bode plots 4)
- MATLAB, "Why the Bode Plot Is Awesome!", YouTube, https://www.youtube.com/watch?v=BGAjnt-C4tI: why only the settled response counts (MATLAB, Why the Bode plot is awesome)
- Åström, K. J. and Murray, R. M. (2021). *Feedback Systems*, 2nd ed., §6.3 "Input/Output Response" (gain, phase, bandwidth, resonant peak), §9.6 "The Bode Plot" (asymptotes, the Q-value $1/(2\zeta)$), §12.2 "Performance Specifications". Free: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-linsys_24Jul2020.pdf, http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-xferfcns_24Jul2020.pdf, http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-loopsyn_24Jul2020.pdf (FBS)
- PX4 Autopilot, parameter files read on GitHub: `src/modules/fw_autotune_attitude_control/fw_a` and `fw_autotune_attitude_control.cpp`, `src/modules/sensors/vehicle_angular_velocity/im` https://github.com/PX4/PX4-Autopilot (PX4)

Other references

- MATLAB, "Using Bode Plots, Part 1: Frequency Response Design Using Sensitivity Functions", https://www.mathworks.com/videos/using-bode-plots-closed-loop-systems-1-of-5-77057.html: the bandwidth rule of thumb, 5 to 10 times the fastest rate to follow

(MATLAB, Using Bode plots 1)

- Virtanen, P. et al. (2020). "SciPy 1.0: fundamental algorithms for scientific computing in Python". *Nature Methods* 17, 261–272: `brentq`, used in the notebook to find the bandwidth

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Gain (amplitude ratio) (G-3021)	How far the output sine swings divided by how far the input sine swings at one rate, such as 1.288 for the heading loop at 1 rad/s; it says how much of a wave of that rate the system passes, often written in decibels.
Phase (phase shift) (G-3022)	How far the output sine is shifted against the input sine at one rate, as a fraction of a period in degrees, negative when the output is late, such as $-14.9°$ for a 0.261 s lag in a 6.28 s period; it says how late a system responds to a wave.
Frequency response (G-3020)	How a stable linear system responds to a sine at each rate ω once its start-up swing has died away: the gain and the phase, equal to the length and angle of its transfer function at $s = j\omega$; it tells how much of a wave of each rate gets through and how late.
Decade (G-3024)	A factor of 10 in frequency, such as 0.25 to 2.5 rad/s; on a log axis every decade has the same width, and Bode-plot slopes are given per decade (-20 dB per decade for one pole).
Bode plot (G-3023)	A pair of plots of a frequency response against the rate on a log scale: the gain in decibels and the phase in degrees; on it the pieces of a system add and the response is sketched with straight lines.
Asymptote (Bode plot) (G-3025)	A straight line a Bode plot's gain or phase curve approaches far from a corner, such as flat at low rates and falling 20 dB per decade per pole at high rates; joining them sketches the plot by hand.

Term	Meaning
Corner frequency (break frequency) (G-3026)	The rate where two asymptotes of a Bode plot meet, one over the time constant for a single pole (0.25 rad/s for the robot's speed plant); there the exact gain is 3.01 dB below the lines and the phase is $-45°$.
Resonance (G-3027)	A large output when a lightly damped system is driven near the rate of its own swing, because each push lands in step with the swing; it makes the heading loop over-turn a weave near 1.92 rad/s.
Resonant peak and resonant frequency (G-3028)	The largest gain of a frequency response, M_r, and the rate where it occurs, ω_r; for a second-order system M_r is about $1/(2\zeta)$ near the natural frequency (2.552 at 1.918 rad/s for $\zeta = 0.2$), and there is no peak when $\zeta \geq 0.707$.
Bandwidth (of a system) (G-3029)	The rate at which a system's gain has fallen to 0.707 of its zero-frequency gain (-3 dB), such as 3.02 rad/s for the first-try heading loop (Kp 0.4, Kd 0.08); it marks the fastest input the system still follows.
PD control (G-2962)	Feedback with a proportional part and a derivative part, such as K_p e + K_d de/dt; it acts like a spring and a damper, so it can bring a frictionless joint or heading to its target without swinging.
Integral action (G-2963)	A part of the command in proportion to the error's running sum (its integral over time); the sum grows while any error remains, so it removes the error a steady load leaves, at the cost of overshoot and, if too large, instability.
Transfer function (control) (G-2999)	The number a linear system multiplies an exponential input e^(st) by, written as a function of s, such as G(s) = 1/(8s + 2) for the robot's speed and $10/s^2$ for its heading; it turns a differential equation into a fraction, so systems can be combined and read by algebra.

Term	Meaning
Step response (G-2981)	The output of a system after a step input, such as the robot's heading after its command jumps to 0.5 rad; its rise time, overshoot, settling time and steady-state error describe how well a loop follows a command.
Tracking (G-2954)	Making an output follow a reference that changes over time, such as a planned speed rising at 0.25 m/s^2; also called the servo problem, it needs the controller to keep up with the change, and a correction from the error alone lags behind.
Pole (G-3000)	A value of s where a transfer function's denominator is zero, so the fraction becomes infinite there, such as -2 (twice) for the PD heading loop; each pole adds a term e^(pt) to every response, so its real part says whether the term decays or grows and its imaginary part how fast it oscillates.
Time constant (G-2988)	The time a first-order system takes to close 63 % of the gap to its final value, m/(K_v + b) = 1 s for the robot's speed loop; it fixes the rise time (about 2.2 time constants) and the 2 % settling time (about 4).
Natural frequency (G-2991)	The rate, in rad/s, at which a second-order system would swing with no damping, $\omega_n = \sqrt{K_p/I}$, 2 rad/s for the heading loop; it sets how fast the response is, and raising it compresses the whole response in time.
Damping ratio (G-2992)	A second-order system's damping compared with the damping that just stops it from swinging, $\zeta = K_d/(2\sqrt{K_p I})$, 0.2 for D gain 0.08 and 1 for 0.4; it alone sets the shape of the response and its overshoot.
Critically damped (G-2995)	A second-order system with damping ratio exactly 1: its characteristic equation has one double real root, so it gives the fastest step response that does not overshoot, such as the turn with K_p = K_d = 0.4 settling in 2.92 s.